Sherif Morsi

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Summary

Concordia Computer Science student seeking AI/ML, software, and quant internships. Trained **PyTorch** models on 5G telemetry at Nokia; builds tool-calling **AI agents**, QLoRA fine-tunes, and leakage-safe forecasting models in tested **Python**.

Education

Concordia University Montreal, QC, Canada
Bachelor of Computer Science *Expected May 2028*

- **Availability:** Winter 2027 (Jan to Apr 2027) for 4, 8, 12, or 16 months
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Linear Algebra, Multivariable Calculus

Experience

AI & Machine Learning Intern Aug 2026 - Sep 2026
Nokia *Cairo, Egypt*

- Trained and evaluated Python and **PyTorch** models on wireless network telemetry to forecast 5G **energy demand**
- Developed **data and ML pipelines** for 5G energy telemetry, to support automated network management and optimization
- Presented findings on wireless networking and 5G-related energy challenges to cross-functional teams of data scientists and engineers; led the project proposal and requirements review

Credit Analyst Job Shadow Aug 2025 - Sep 2025
Qatar National Bank (QNB) *Cairo, Egypt*

- Shadowed credit analysts reviewing SME financial records as part of the bank's credit-review process
- Built **Python** utilities and **SQL** queries to preprocess and validate SME financial records, improving the data quality and reliability of credit-review data
- Tested and documented these scripts and fixes with the team so they stayed maintainable and easy to debug

Projects

Orbit (Agentic AI, Multi-Agent Tool Calling) | *Python, PyTorch, QLoRA, Claude API* | **GitHub** 2026

- Built a planner, coder, test runner, and reviewer; the reviewer rejects a patch that fails the hidden tests or edits the tests
- On code-eval-v2 (32 frozen bug-fix tasks, claude-haiku-5-5, scored once), the team passed **29/32** and a single agent **27/32**; estimated cost **$0.025**. The paired 95% interval on pass@1 is **0.000 to 0.15625** and includes zero, so the team is not significantly better
- QLoRA-tuned **Qwen2.5-1.5B** on harness v3: schema-valid rate **0.333 to 0.764** (95% CI 0.264 to 0.607, above zero). Task success was **4/12 vs 2/12** (95% CI −0.167 to 0.500, includes zero)

Time-Series Foundation Model (Transformers, Forecasting) | *Python, PyTorch, pytest, GitHub Actions* | **GitHub** 2026

- Pretrained a **10.1M-parameter** decoder-only transformer from scratch. On the frozen test (patch_s2, one seed, 97 configurations), CRPS was **0.688**× upstream AutoARIMA, 95% CI **[0.624, 0.757]**, better
- Validation CRPS was **0.605**× **Seasonal Naive** (16 configurations, one seed). A content audit flagged **801 of 447,229** series and excluded them; the corpus kept is **443,258** series across **77** datasets

World Shock Lab (Econometrics, Macro Shocks) | *Python, PyTorch, DuckDB, TypeScript* | **GitHub** | **Live Site** 2026

- Estimated Fed, Bank of Canada, and OPEC surprise effects on **53** assets: a 10 bp Fed surprise moves the TSX **−1.10%** (90% band −1.59 to −0.61); the same-size BoC surprise moves it **−0.19%**
- Replicated **11 of 13** published findings on sign and 90% band. Quantile gradient boosting cut 5% pinball loss by **0.112** versus a Gaussian projection (90% CI −0.228 to −0.010). Slowest throttled phone view: **2.703 s**

ATLAS (Production Networks) | *Python, PyTorch, FastAPI* | **GitHub** 2026

- Traced a hypothetical cost shock through **405** country-sector nodes (local median **11.0 ms**). On the one held-out year, ridge test RMSE **14.00** was below every saved graph and no-graph checkpoint

Certifications & Affiliations

Anthropic: Building with the Claude API, Claude Code 101, Claude 101 | **IBM:** Machine Learning with Python
Quantitative Research and Competitions Club (QUARCC) | *Member, Concordia University*: Exploring machine learning concepts, statistical modeling, and quantitative research through data-driven problem-solving

Technical Skills

Languages: **Python**, **SQL**, C++, **Java**, C, R, JavaScript, TypeScript, Bash, MATLAB
Machine Learning: **PyTorch**, TensorFlow, **scikit-learn**, Deep Learning, Supervised/Unsupervised Learning, Regression, **Transformers**, **Time-Series Forecasting**
ML Methods: **Feature Engineering**, **Cross-Validation**, Transfer Learning, Computer Vision, NLP, **Model Evaluation**, Mixed Precision
AI / LLMs: **LLMs**, **AI Agents**, **Tool Calling**, **Prompt Engineering**, QLoRA, Distillation, Claude API, Hugging Face
Quant / Risk: **Econometrics**, Event Studies, Monte Carlo, Probabilistic Forecasting, **CVaR**, **Portfolio Optimization**, Walk-Forward Validation
Data / Libraries: **Pandas**, **NumPy**, SciPy, Matplotlib, OpenCV, **FastAPI**, **DuckDB**, **SQLite**, Parquet, cvxpy, statsmodels
Developer Tools: **Git**, **GitHub Actions (CI/CD)**, **pytest**, Playwright, Jupyter, Colab, Claude Code, Cursor, Codex, VS Code
Office / BI: Excel (Advanced), Power BI, PowerPoint, Microsoft 365